NORTH SOUTH UNIVERSITY

Project Proposal

Multilingual Speech-to-Speech Voice Cloning via Knowledge Distillation over Large Voice Clone Models to Tiny On-Device Model

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Section 1

Abstract

Zero-shot voice cloning, converting any speaker's voice without prior training on that speaker. has improved dramatically with large generative models, but these models are far too large and slow to run on smartphones or edge devices. Worse, most high-quality voice cloning systems only work well in English, making multilingual and cross-lingual speaker transfer an open problem at practical scales.

We propose a knowledge distillation framework that compresses large multilingual voice cloning models into tiny, on-device student networks around 50-100 million parameters, all without using any text, keeping the pipeline purely speech-in, speech-out. A large teacher model with a multilingual speech encoder guides a lightweight student consisting of a causal conformer, a shared cross-lingual speaker encoder, and a compact HiFi-GAN vocoder, trained jointly with speaker similarity, pitch, and cycle-consistency losses. The resulting model should run in real-time under 100ms latency on mobile hardware and supports cross-lingual voice cloning across five or more languages. Experiments on FLEURS and EMIME will show our distilled model retains competitive speaker similarity and naturalness compared to teacher-scale systems establishing the first knowledge distillation pipeline for multilingual, on-device, text-free voice cloning.

Dataset Distribution & Benchmark Sets

5.1 Training Datasets

Dataset	Languages	Hours	Speakers	License	Used In
LibriTTS (Zen et al., 2019)	English	585h	2,456	CC BY 4.0	StreamVC, Seed-VC, FreeVC, RT-VC
Multilingual LibriSpeech (MLS)	8 (EN, DE, FR, ES, IT, PT, NL, PL)	50,000 h+	Thousands	CC BY 4.0	CosyVoice, YourTTS, MMS
CommonVoice v15 (Mozilla)	100+ languages	Varied	Crowdsourced	CC0 (Public Domain)	MMS, Whisper fine-tuning
VoxPopuli (Meta, 2021)	23 EU languages	400,000h	Thousands	CC0 / CC BY	Cross-lingual SSL pretraining
AISHELL-3 (Chinese)	Mandarin	85h	218	CC BY-NC-ND	Chinese VC evaluation

VCTK Corpus (CSTR)	English (varied accents)	44h	110	Open (CSTR)	StreamVC, YourTTS, QuickVC eval

5.2 Evaluation / Benchmark Datasets

Dataset / Benchmark	Languages	Purpose	Why This Project Uses It
VCC 2020 (Voice Conversion Challenge)	English	Standardized VC eval with human MOS scores	Gold standard benchmark — every VC paper uses it
VCTK Test Set (11 unseen speakers)	English	Speaker similarity and WER evaluation	Most common VC test set; enables fair comparison with StreamVC and RT-VC
FLEURS (Google, 2022)	102 languages	Cross-lingual speech quality evaluation	Best multilingual benchmark; enables per-language quality reporting
MLS Test Split (Pratap et al.)	8 languages	Multilingual VC output intelligibility	Large-scale multilingual benchmark with clean test speakers
AISHELL-3 Test Split	Mandarin	Chinese voice cloning fidelity	Key non-European language; tests whether model generalizes beyond Latin-script languages

5.3 Evaluation Metrics

Metric	What It Measures	How to Compute	Target Goal
SECS — Speaker Embedding Cosine Similarity	Does the output sound like the target speaker?	Cosine similarity between WeSpeaker/ECAPA embeddings of output and reference	> 0.82 on VCTK
WER / CER — Word / Character Error Rate	Are the words still understandable?	Transcribe output with Whisper-Large; compare to original text	< 5% WER (English), <

			8% per language
DNSMOS / MOS-P.835	Does it sound natural to human ears?	Microsoft's non-intrusive MOS predictor (automatic)	> 3.5 out of 5
MCD — Mel Cepstral Distortion	How different is the spectrum from the teacher output?	Compute MCD between student mel and teacher mel on same input	< 4.5 dB
RTF — Real-Time Factor	Is it fast enough for live use?	RTF = processing_time / audio_duration; RTF < 1 means real-time	< 0.3 on RTX 5060Ti
Params (#M)	How big is the model?	Count all trainable parameters	< 50M total
Cross-lingual SECS	Does voice identity survive language change?	SECS when source and target are in different languages	> 0.75 (new benchmark)

Multiple Proposed Methodology (Subject to be updated)

Methodology 1: Two-Stage Progressive Distillation

Train the student in two stages. First, train just the content encoder (the part that extracts linguistic information from speech) to mimic a large multilingual encoder. Then, train the full student model to produce voice conversions that look like what the teacher produces.

Student Model Architecture Target: 35 - 45M Parameters

Component	**What It Does**	**Architecture**	**Params**
Causal Content Encoder	Extracts words / phonemes from speech without revealing speaker identity	6-layer causal Conv-Transformer with causal masking (processes left-to-right only for streaming)	~8M
Multilingual Speaker Encoder	Creates a compact voice fingerprint (embedding) from 3–5 seconds of reference audio	ECAPA-TDNN with multilingual training on MLS + AISHELL-3	~10M
Language ID Module	Detects what language is being spoken	Simple 3-layer CNN classifier trained on FLEURS	~1M

Mel Decoder	Combines content + speaker + language info to generate the target speech spectrogram	Non-autoregressive Transformer decoder with FiLM conditioning for language	~12M
Vocoder (HiFi-GAN Lite)	Converts spectrogram to actual audio waveform	HiFi-GAN with reduced channels — trained separately first	~12M

Methodology 2: Consistency Distillation from Diffusion Teacher

Seed-VC uses a diffusion model that runs 10–50 steps to produce one audio output. Consistency distillation is a technique (from LCM: Latent Consistency Models, NeurIPS 2023) that collapses those many steps into just 1–4 steps, without retraining from scratch. The resulting student is both fast AND inherits Seed-VC's quality. A normal diffusion model needs 50 steps × network cost per step = very slow. If we train a consistency model that jumps directly from noisy audio to clean audio in one step, it runs 10–50x faster while producing nearly the same result. This is not just compression, it is a fundamentally different inference method.
Because Seed-VC is English-focused, CosyVoice is used as a second teacher for non-English languages. During training, each batch sample is assigned to the correct teacher based on its source language. The consistency loss is computed against that teacher's output.

Architecture

- Teacher: Seed-VC U-DiT (full, ~300M params, 50-step flow matching) — kept frozen
- Student: Reduced U-DiT — 50% fewer transformer layers and half the attention heads → target ~45M params
- Modification: Add causal masking to all attention layers so the model can run in streaming mode
- Multilingual: Replace the content encoder with a distilled Whisper-Tiny-Causal (~8M params)

Methodology 3: Multi-Teacher Ensemble Distillation with Language Routing

No single teacher model is best at everything. Seed-VC is the best at English quality. CosyVoice is the best at multilingual coverage. EZ-VC handles Indian languages. This methodology uses all of them as teachers simultaneously, routing each training sample to the teacher that is best suited for that language. The student learns from all teachers at once, becoming better than any single teacher alone at the task of multilingual on-device VC.

The Language Router

A small language routing module (2M params, based on a frozen language ID classifier trained on FLEURS) reads the source audio and outputs a language probability distribution. This distribution is used to:

- Select which teacher generates the ground-truth for that training sample
- Weight the distillation loss appropriately (low-resource languages get higher weight to prevent English dominance)
- Activate the correct FiLM (Feature-wise Linear Modulation) conditioning layer in the decoder for that language

Full Architecture Target: ~46M Parameters

Component	Architecture	Params	Notes
Causal Content Encoder	Distilled Whisper-Tiny (causal, 6-layer)	~8M	Multilingual by design; processes left-to-right for streaming
Language Router	3-layer CNN + softmax over 15 language classes	~2M	Frozen after initial training; activates correct FiLM layer
Multilingual Speaker Encoder	ECAPA-TDNN trained on MLS + AISHELL-3 + CommonVoice	~10M	Language-agnostic speaker embedding
Language-Conditioned Decoder	6-layer Transformer with per-language FiLM layers	~16M	Each language has 0.5M FiLM params; shared backbone
HiFi-GAN Lite Vocoder	Reduced HiFi-GAN (3 upsampling blocks)	~10M	Trained first separately, then fine-tuned jointly

Computation Power

Primilarily we will be using NVIDIA RTX 3060ti 8GB GPU & NVIDIA RTX 5060ti 16GB GPU for training purpose on our local devices. Furthermore, we will use Google Colab T4 GPU whenever needed.

References

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